

# Dat Lam

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## EDUCATION

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### Carleton University

Ottawa, ON

*Bachelor of Science in Computer Science (Honours)*

- Relevant Coursework: Operating Systems, Design and Analysis of Algorithms, Database Management Systems, Discrete Structures, Fundamentals of Web Applications.

## TECHNICAL SKILLS

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**Languages:** Java, C++, C#, HTML, CSS, JavaScript, SQL.

**Frameworks & Libraries:** React, Node.js, Spring Boot, PostgreSQL.

**Developer Tools:** Git, GitHub, Linux, GDB.

## PROJECTS

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### Nyctophobia

Godot

*A puzzle Hack and Slash game*

- Architected diverse enemy archetypes with dynamic difficulty scaling, progressively amplifying monster attributes over time. Implemented a procedural maze generation algorithm that adjusts structural complexity based on player progression metrics.
- Engineered a tiered loot system featuring three chest rarities, each with weighted probability distributions that increase powerful buff drop rates at higher price points, incentivizing strategic resource allocation.
- Developed shortest-path enemy navigation using A\* pathfinding to dynamically route enemies through the maze toward the player. Designed original character assets and built an autonomous pet AI companion with priority-based target selection and state-driven behavior logic.

### Cave Diver

Godot

*A dungeon crawler game*

- Designed and balanced enemy encounter systems with algorithmic wave-based spawn patterns, ensuring consistent difficulty curves and challenge progression across dungeon levels.
- Built an intuitive HUD and menu system with clear visual hierarchy and responsive feedback. Implemented character ability systems and coordinated cross-discipline tasks across design, art, and programming pipelines to deliver milestones on schedule.

### Personal Portfolio

React, JavaScript

*Personal website*

- Designed and developed a responsive single-page portfolio application featuring smooth scroll navigation, dynamic project filtering, and a mobile-first layout optimized for all viewports.
- Implemented modular component architecture using React functional components and hooks (useState, useEffect) for efficient state management. Styled with Tailwind CSS utility classes, delivering consistent theming and adaptive layouts across desktop and mobile devices.
- Styled interface using Tailwind CSS with custom dark green and amber color scheme, ensuring consistent theming and responsive layouts across all device sizes.